column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
$\therefore$ The condition that for each column having a leading entry, all positions below it must contain 0 is True.
So, the final matrix is in echelon form. □
If $a_1 + a_2/8 - 19a_3/8 \neq 0$, then by Theorem 2, the system of equations corresponding to the augmented matrix $\begin{bmatrix} 3 & 1 & a_1 \\ 5 & 7 & a_2 \\ 1 & 3 & a_3 \end{bmatrix}$ has no solution.

$\therefore a \notin \text{Span}\left\{\begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 7 \\ 3 \end{bmatrix}\right\}$ when $8a_1 + a_2 - 19a_3 \neq 0$.

$\therefore$ T is not onto □

Practice Problems

1. Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be the transformation that first performs a horizontal shear that maps e_2 into $e_2 - .5e_1$ (but leaves e_1 unchanged) and then reflects the result through the x_2-axis. Assuming that T is linear, find its standard matrix. (Good Prob)

Solution: **Transformation 1:** $e_1 \to e_1$. $\begin{bmatrix} 0 \\ 1 \end{bmatrix} = e_2 \to e_2 - 0.5e_1$
$= \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 0.5\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
$= \begin{bmatrix} 0-0.5 \\ 1-0 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 1 \end{bmatrix}$

Standard matrix
$\begin{bmatrix} 1 & -0.5 \\ 0 & 1 \end{bmatrix}$

Transformation 2: $e_1 \to (-1) \cdot e_1$

The vector $e_2 - 0.5e_1$ transforms to $e_2 + 0.5e_1$. $\therefore T(e_1) = -e_1$ and $T(e_2) = e_2 + 0.5e_1$

Thus the standard matrix of T is $\begin{bmatrix} T(e_1) & T(e_2) \end{bmatrix} = \begin{bmatrix} -1 & 0.5 \\ 0 & 1 \end{bmatrix}$

$\therefore$ The final standard matrix is $\begin{bmatrix} -1 & 0.5 \\ 0 & 1 \end{bmatrix}$ □